

# Zane Perry

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## EDUCATION

University of Colorado Boulder

Boulder, CO

**M.S. Applied Mathematics**

August 2025 - Expected May 2026

Dynamics & Numerical Modeling, Applied Data Science & Stochastic Processes, Machine Learning; Cumulative GPA: 4.0

**B.S. in Computer Science, B.S. in Applied Mathematics**

August 2021 - May 2025

Summa Cum Laude, Dean's List Fall 2021- Spring 2025, Cumulative GPA: 4.0

## Skills

**Languages:** Python, Java, TypeScript/JavaScript, C/C++, SQL, MATLAB, Bash

**Tools:** React, Git, Docker, Numpy, Scipy, PyTorch, Pandas, Sci-Kit Learn, CUDA

**Math:** ODE/PDE Methods, Numerical Analysis, Machine/Deep Learning, Signal Processing

## WORK EXPERIENCE

**CU Electrical Engineering Department - Genetic Logic Lab**

Boulder, CO

Research Assistant, Software Developer

June 2022 – Present

- Design, Implement, and Test frontend/backend frameworks and endpoints using React, Spring, Java, and Javascript for a Genetic info repository, enhancing data accessibility and system reliability. Optimize legacy code for future scalability.
- Execute full stack development based on specified user requirements. Manage deployment on Azure and Kubernetes.
- Mentor an undergraduate student on the software development life cycle.

**Handshake MOVE Fellowship**

Remote

AI Trainer - Math Expert, Reviewer (Contract)

May 2025 – December 2025

- Developed 100 complex mathematical prompts to assess and improve the performance of large language models.
- Analyzed LLM outputs for mathematical rigor, enhancing model complexity and outputs.
- Audited other fellows prompts and responses to address complexity, analyze reasoning, and suggest improvements.

**Space Environment Technologies**

Denver, CO

Radiation Division Computer Science Intern

May 2024 – August 2024

- Automated data upload/analysis processes with APIs and file systems, to reduce manual processing time by 95%.
- Worked with specialized atmospheric radiation models in order to enhance understanding of space radiation effects.
- Designed and built Agile front-end user apps for flight planning and radiation risk analysis.

**CU Computer Science Department**

Boulder, CO

Data Science Course Assistant for 300 students

August 2023 - December 2023

- Reviewed homework assignments and lecture topics to recommend improvements for the benefit of student learning.
- Held office hours to review concepts, including practical coding applications and theoretical mathematical derivations.

## AWARDS AND LEADERSHIP

- National Merit Scholar, 2021.
- University of Colorado, Boulder Engineering Scholarship, Fall 2021-Spring 2025.
- University of Colorado, Boulder Chancellor's Recognition Award, Spring 2025.
- President, Theta Tau Professional Engineering Fraternity Eta Gamma Chapter, (100 members, 300 alumni), 2022-2025.
- Chair/Webmaster, Engineering Excellence Fund, (responsible for \$500k in engineering grants), 2021-2023.
- Writer/Webmaster, Colorado Engineering Magazine, (Distributed to 40k students/alumni), 2021-2025.

# TECHNICAL PROJECT PORTFOLIO – UNIVERSITY OF COLORADO, BOULDER

## Synbiohub 2/3, A Standards Based Genetic Design Repository

Spring 2022 - Present

- **Motivation:** Modernize a tightly coupled, legacy genetic design repository that struggled to scale and was difficult to evolve alongside SBOL and triplestore standards and modern web UX expectations.
- **Approach:** Built a modular plugin interface to decouple extensions from the core platform, and designed a cleaner frontend/backend separation around REST APIs.
- **Tech and Methods:** React/Redux, Java/Spring, REST APIs, PostgreSQL & SPARQL, Docker/Kubernetes/Azure.

## Turing Patterns and Phase Separation in Morphogenesis

December 2025

- **Motivation:** Investigated why reaction–diffusion Turing systems and phase-separation models can produce visually similar biological patterns (stripes/spots), and what that implies for morphogenesis modeling.
- **Approach:** Derived the Brusselator and Ohta–Kawasaki PDEs from conservation/flux/free-energy assumptions, then performed linear stability and Fourier-mode analysis to obtain dispersion relations, unstable wavenumber bands, nonlinear amplitudes, bifurcations, and numerical simulations.
- **Tech and Methods:** Dynamical Systems, PDEs, Bifurcations, Numerical Simulation.

## Senior Capstone - Festo AI Assisted Rapid Prototyper for DevOps

April 2025

- **Motivation:** Speed up DevOps prototyping by turning uploaded Epics/User Stories into a runnable starter app, reducing time spent on initial scaffolding and wiring.
- **Approach:** Built a React workflow to upload, generate, edit/view, and export code, integrating an LLM-backed Python service and an in-browser dev environment for inspecting and iterating on generated code and file structure. Worked with a group of 6 students through the entire process overseen by a corporate sponsor.
- **Tech and Methods:** React/Redux, AWS DynamoDB, Generative AI, Chain of Thought Prompting.

## Advanced Music Recommendation using Deep Learning and Digital Signal Processing

April 2025

- **Motivation:** Explore music-first recommendation (audio content over metadata/behavior graphs) to better handle personalization and cold-start settings where historical signals are sparse.
- **Approach:** Built an end-to-end pipeline that converts raw audio into DSP feature representations (MFCC, STFT, DWT), then trains single-modal and multimodal neural models for binary user–song compatibility.
- **Tech and Methods:** Hybrid CNN/RNN/LSTM, Signal Transforms, Hyperparameter Analysis, PyTorch, Sci-Kit, TorchAudio.

## Boundary Integral Equation Ordinary/Partial Differential Equation Solver

April 2024

- **Motivation:** Avoid instability/ill-conditioning from mesh-based derivative discretization by reformulating boundary value problems as integral equations where numerical error is driven primarily by quadrature accuracy.
- **Approach:** Derived Green's-function solutions for 1D Sturm–Liouville BVPs and extended to 2D Laplace problems using single/double-layer potentials, reducing PDE solves to a boundary-only Fredholm integral equation for an unknown density.
- **Tech and Methods:** Numerical Quadrature, Boundary Discretization, Irregular Domains, Dense Solvers, Numpy, Scipy.

## Airfoil Fluid Simulation Using Conformal Mapping

April 2023

- **Motivation:** Model lift-generating flow over an airfoil without full CFD by using ideal potential flow + complex analysis to translate simple geometries into realistic wing cross-sections.
- **Approach:** Constructed the complex velocity potential by superimposing uniform/source–sink/doublet/vortex flows, enforced the Kutta condition, then applied the Joukowski conformal map to transform cylinder flow into a cambered airfoil and compute circulation/lift (Kutta–Joukowski).
- **Tech and Methods:** Complex Variables, PDEs, Numerical Visualizations, MATLAB.

## Signal Noise Reducer Using Numerical Matrix Factoring

December 2023

- **Motivation:** Denoise signals efficiently by extracting a low-rank structure without relying on full SVD, which becomes expensive on large matrices.
- **Approach:** Implemented rank-revealing QR with column pivoting (and compared Gram–Schmidt, Householder, Givens variants) to estimate numerical rank and build low-rank approximations.
- **Tech and Methods:** Subspace Methods, Signal Processing, Python, Numpy, Numerical Linear Algebra.

## Image Compression Using Discrete Wavelet Transforms

December 2022

- **Motivation:** Reduce image storage size while preserving perceptual quality by exploiting the fact that most images concentrate information in low-frequency structure.
- **Approach:** Implemented a 2D Haar DWT as an averaging orthogonal transform, then applied coefficient thresholding to induce sparsity and reconstruct the image via the inverse transform; compared against SVD compression as a numerical baseline.
- **Tech and Methods:** MATLAB, Signal Processing, Numerical Linear Algebra, Computer storage.

For Full Project Descriptions and Artifacts visit: <https://zane-perry.com/projects>